Land Management System

Entity · Attribute · Relation Datasheet + MySQL DDL

Database Design Reference | Version 2.0

Document overview

This document covers the complete database design for the Land Management System, including 15 entities, all attributes with data types and constraints, entity-relationship table, and full MySQL DDL including triggers, events, indexes, and views.

Tag	Meaning	Entities
ADMIN	Admin-only entity	ADMIN
USER	Citizen-facing entity	USER
CORE	Core land system entity	LAND, TRANSACTION, SURVEY, etc.
NEW	Upgraded / new feature entity	BILLBOARD_PAYMENT, DISPUTE (escalation), DOCUMENT (expiry), AUDIT_LOG

Section 1 — Entities and attributes

ADMIN [ADMIN]

System administrators who manage, approve, and operate all land-related records. Separated from USER for role-based access control.

Attribute	Data type	Constraint	Description
admin_id	INT	PK, AUTO_INCREMENT	Unique admin identifier
full_name	VARCHAR(100)	NOT NULL	Administrator full name
email	VARCHAR(150)	UNIQUE, NOT NULL	Login email address
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
role	ENUM('superadmin','officer','surveyor')	NOT NULL	Admin role type
last_login	DATETIME	DEFAULT NULL	Timestamp of last login
is_active	TINYINT(1)	DEFAULT 1	Account active flag
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation time

USER [USER]

Registered citizens who own land, raise disputes, request work orders, and pay taxes. Separate from ADMIN for clean permission boundaries.

Attribute	Data type	Constraint	Description
user_id	INT	PK, AUTO_INCREMENT	Unique user identifier
full_name	VARCHAR(100)	NOT NULL	Citizen full name
nid_number	VARCHAR(30)	UNIQUE, NOT NULL	National ID number
email	VARCHAR(150)	UNIQUE, DEFAULT NULL	Contact email
phone	VARCHAR(20)	NOT NULL	Contact phone number
address	TEXT	DEFAULT NULL	Residential address
is_verified	TINYINT(1)	DEFAULT 0	NID verification status
registered_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Registration timestamp

LAND [CORE]

Central entity. Every other entity references LAND. Stores parcel details, location, current ownership, estimated value (auto-computed from transactions).

Attribute	Data type	Constraint	Description
land_id	INT	PK, AUTO_INCREMENT	Unique parcel identifier
parcel_number	VARCHAR(50)	UNIQUE, NOT NULL	Official parcel/dag number
area_sqft	DECIMAL(12,2)	NOT NULL	Land area in square feet
land_type	ENUM('residential','agricultural','commercial','industrial','government')	NOT NULL	Classification of land use
mouza	VARCHAR(100)	NOT NULL	Mouza name
upazila	VARCHAR(100)	NOT NULL	Upazila name
district	VARCHAR(100)	NOT NULL	District name
current_status	ENUM('active','disputed','transferred','government')	DEFAULT 'active'	Current land status
current_estimated_value	DECIMAL(15,2)	DEFAULT 0.00	Auto-computed from transaction history
registered_owner_id	INT	FK → USER(user_id)	Current registered owner
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Record creation time

LAND_TRANSACTION [CORE]

Every ownership transfer. Includes a chain_valid flag verified by stored procedure to detect broken ownership chains.

Attribute	Data type	Constraint	Description
transaction_id	INT	PK, AUTO_INCREMENT	Unique transaction ID
land_id	INT	FK → LAND(land_id)	Parcel involved
from_user_id	INT	FK → USER(user_id)	Previous owner
to_user_id	INT	FK → USER(user_id)	New owner
transaction_type	ENUM('sale','inheritance','gift','court_order')	NOT NULL	Nature of transfer
transaction_date	DATE	NOT NULL	Date of transfer
amount	DECIMAL(15,2)	DEFAULT 0.00	Transaction value in BDT
deed_number	VARCHAR(100)	UNIQUE, NOT NULL	Official deed reference
chain_valid	TINYINT(1)	DEFAULT 1	Ownership chain integrity flag

Attribute	Data type	Constraint	Description
approved_by	INT	FK → ADMIN(admin_id)	Approving admin
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Record timestamp

LAND_HISTORY [CORE]

Immutable event log per parcel — surveys, transactions, status changes, legal events.

Attribute	Data type	Constraint	Description
history_id	INT	PK, AUTO_INCREMENT	Unique history record
land_id	INT	FK → LAND(land_id)	Related parcel
event_type	ENUM('survey','transaction','dispute_filed','status_change','billboard_placed','work_order')	NOT NULL	Category of event
description	TEXT	NOT NULL	Detailed event notes
event_date	DATE	NOT NULL	Date of event
recorded_by_admin	INT	FK → ADMIN(admin_id)	Admin who logged the event

LAND_SURVEY [CORE]

Survey records with surveyor, method, and approval. Each survey captures land type at time of survey for historical accuracy.

Attribute	Data type	Constraint	Description
survey_id	INT	PK, AUTO_INCREMENT	Unique survey record
land_id	INT	FK → LAND(land_id)	Surveyed parcel
survey_date	DATE	NOT NULL	Date of survey
surveyor_admin_id	INT	FK → ADMIN(admin_id)	Admin who conducted survey
land_type_at_survey	VARCHAR(50)	NOT NULL	Land classification at time of survey
survey_method	ENUM('manual','digital','satellite')	NOT NULL	Survey methodology
notes	TEXT	DEFAULT NULL	Survey remarks
approved_by	INT	FK → ADMIN(admin_id)	Admin who approved survey

INFRASTRUCTURE [CORE]

Physical assets on a land parcel — buildings, roads, drainage, bridges.

Attribute	Data type	Constraint	Description
asset_id	INT	PK, AUTO_INCREMENT	Unique asset ID
land_id	INT	FK → LAND(land_id)	Associated parcel
asset_name	VARCHAR(150)	NOT NULL	Name of the asset
asset_type	ENUM('building','road','drainage','bridge','utility')	NOT NULL	Category of infrastructure
construction_date	DATE	DEFAULT NULL	Date of construction
condition	ENUM('excellent','good','fair','poor','critical')	DEFAULT 'good'	Current physical condition
owner_entity	VARCHAR(100)	DEFAULT NULL	Owning government body

MAINTENANCE [CORE]

Maintenance records for infrastructure assets.

Attribute	Data type	Constraint	Description
maintenance_id	INT	PK, AUTO_INCREMENT	Unique maintenance record
asset_id	INT	FK → INFRASTRUCTURE(asset_id)	Asset being maintained
maintenance_date	DATE	NOT NULL	Date of maintenance
maintenance_type	ENUM('routine','emergency','upgrade','demolition')	NOT NULL	Type of work
cost	DECIMAL(12,2)	DEFAULT 0.00	Total cost in BDT
assigned_admin_id	INT	FK → ADMIN(admin_id)	Responsible admin
notes	TEXT	DEFAULT NULL	Maintenance notes
next_scheduled_date	DATE	DEFAULT NULL	Next maintenance due date

WORK_ORDER [CORE]

Government work orders for land-related activities. Users request; admins are assigned.

Attribute	Data type	Constraint	Description
work_order_id	INT	PK, AUTO_INCREMENT	Unique work order ID
land_id	INT	FK → LAND(land_id)	Target parcel
work_type	ENUM('road_construction','survey','demolition','utility_install','inspection')	NOT NULL	Nature of work
requested_by_user	INT	FK → USER(user_id)	Citizen who requested
assigned_admin	INT	FK → ADMIN(admin_id)	Admin responsible
status	ENUM('pending','in_progress','completed','cancelled')	DEFAULT 'pending'	Work order status
start_date	DATE	DEFAULT NULL	Planned start date
end_date	DATE	DEFAULT NULL	Planned end date
notes	TEXT	DEFAULT NULL	Additional notes

BILLBOARD [CORE]

Billboard licenses on land parcels. Linked to BILLBOARD_PAYMENT for revenue tracking.

Attribute	Data type	Constraint	Description
billboard_id	INT	PK, AUTO_INCREMENT	Unique billboard ID
land_id	INT	FK → LAND(land_id)	Land parcel hosting billboard
owner_user_id	INT	FK → USER(user_id)	License holder
license_number	VARCHAR(80)	UNIQUE, NOT NULL	Official license reference
size_sqft	DECIMAL(8,2)	NOT NULL	Billboard size in sq ft
installation_date	DATE	NOT NULL	Date installed
expiry_date	DATE	NOT NULL	License expiry date
annual_fee	DECIMAL(10,2)	NOT NULL	Annual license fee in BDT
status	ENUM('active','expired','suspended','removed')	DEFAULT 'active'	Current license status

BILLBOARD_PAYMENT [NEW]

NEW — Tracks installment payments per billboard license. Enables revenue reporting and delinquency detection.

Attribute	Data type	Constraint	Description
payment_id	INT	PK, AUTO_INCREMENT	Unique payment record
billboard_id	INT	FK → BILLBOARD(billboard_id)	Related billboard
payment_date	DATE	NOT NULL	Date payment received
amount_paid	DECIMAL(10,2)	NOT NULL	Amount paid in BDT
payment_method	ENUM('cash','bank_transfer','online')	NOT NULL	Payment channel
status	ENUM('paid','pending','overdue')	DEFAULT 'pending'	Payment status
received_by_admin	INT	FK → ADMIN(admin_id)	Admin who received payment

TAX [CORE]

Land tax per parcel per financial year. Value derived from current_estimated_value in LAND.

Attribute	Data type	Constraint	Description
tax_id	INT	PK, AUTO_INCREMENT	Unique tax record
land_id	INT	FK → LAND(land_id)	Taxed parcel
owner_user_id	INT	FK → USER(user_id)	Taxpayer
financial_year	VARCHAR(10)	NOT NULL	e.g. 2024-2025
assessed_value	DECIMAL(15,2)	NOT NULL	Base value for tax calculation
tax_rate	DECIMAL(5,4)	NOT NULL	Applicable tax rate (e.g. 0.0150)
tax_amount	DECIMAL(12,2)	NOT NULL	Computed tax (assessed_value × tax_rate)
paid_amount	DECIMAL(12,2)	DEFAULT 0.00	Amount paid so far
due_date	DATE	NOT NULL	Payment deadline
status	ENUM('unpaid','partial','paid','overdue')	DEFAULT 'unpaid'	Payment status
calculated_by	INT	FK → ADMIN(admin_id)	Admin who generated tax

DISPUTE [NEW]

NEW — Includes escalation_level (1–3) auto-incremented by trigger if no status change within 30 days.

Attribute	Data type	Constraint	Description
dispute_id	INT	PK, AUTO_INCREMENT	Unique dispute ID
land_id	INT	FK → LAND(land_id)	Disputed parcel
raised_by_user	INT	FK → USER(user_id)	Citizen who filed
opponent_party	VARCHAR(150)	DEFAULT NULL	Name of opposing party
dispute_type	ENUM('ownership','boundary','mutation','encroachment','tax')	NOT NULL	Category of dispute
status	ENUM('filed','under_review','resolved','escalated','dismissed')	DEFAULT 'filed'	Current status
escalation_level	TINYINT	DEFAULT 1	Auto-escalates 1→2→3 after 30 days idle
filed_date	DATE	NOT NULL	Date dispute was filed
last_updated	DATETIME	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Auto-updated on any change
resolution_date	DATE	DEFAULT NULL	Date resolved (if resolved)
resolution_notes	TEXT	DEFAULT NULL	Resolution summary
handled_by_admin	INT	FK → ADMIN(admin_id)	Assigned admin officer

DOCUMENT_REGISTRATION [NEW]

NEW — Documents carry expiry_date and doc_status for proactive lifecycle tracking of deeds, survey certs, tax clearances.

Attribute	Data type	Constraint	Description
document_id	INT	PK, AUTO_INCREMENT	Unique document ID
land_id	INT	FK → LAND(land_id)	Associated parcel
document_type	ENUM('deed','survey_certificate','tax_clearance','court_order','mutation_order','other')	NOT NULL	Document category
title	VARCHAR(200)	NOT NULL	Document title
file_reference	VARCHAR(255)	DEFAULT NULL	Physical/digital file path
registration_date	DATE	NOT NULL	Date registered

Attribute	Data type	Constraint	Description
expiry_date	DATE	DEFAULT NULL	Expiry date if applicable
doc_status	ENUM('valid','expiring_soon','expired','revoked')	DEFAULT 'valid'	Lifecycle status
verified	TINYINT(1)	DEFAULT 0	Admin verification flag
registered_by_admin	INT	FK → ADMIN(admin_id)	Admin who registered document

AUDIT_LOG [NEW]

NEW — Immutable log of every INSERT/UPDATE/DELETE on critical tables. Makes the system legally defensible. Populated via MySQL triggers.

Attribute	Data type	Constraint	Description
log_id	INT	PK, AUTO_INCREMENT	Unique log entry
table_name	VARCHAR(80)	NOT NULL	Affected table name
operation	ENUM('INSERT','UPDATE','DELETE')	NOT NULL	Operation type
record_id	INT	NOT NULL	PK of affected record
admin_id	INT	FK → ADMIN(admin_id), DEFAULT NULL	Admin who triggered change
user_id	INT	FK → USER(user_id), DEFAULT NULL	User who triggered change
old_value	TEXT	DEFAULT NULL	Previous value (JSON)
new_value	TEXT	DEFAULT NULL	New value (JSON)
performed_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp of operation

Section 2 — Entity relationships

From entity	To entity	Cardinality	Relationship meaning
ADMIN	LAND_TRANSACTION	1 to Many	admin approves multiple transactions
ADMIN	LAND_HISTORY	1 to Many	admin records land history events
ADMIN	LAND_SURVEY	1 to Many	admin conducts and approves surveys
ADMIN	MAINTENANCE	1 to Many	admin assigned to maintenance tasks
ADMIN	WORK_ORDER	1 to Many	admin assigned to work orders
ADMIN	TAX	1 to Many	admin calculates tax records
ADMIN	DISPUTE	1 to Many	admin handles disputes
ADMIN	DOCUMENT_REGISTRATION	1 to Many	admin registers documents
ADMIN	BILLBOARD_PAYMENT	1 to Many	admin receives billboard payments
ADMIN	AUDIT_LOG	1 to Many	admin actions logged in audit
USER	LAND	1 to Many	user owns multiple parcels
USER	LAND_TRANSACTION	1 to Many	user buys or sells land
USER	WORK_ORDER	1 to Many	user requests work orders
USER	BILLBOARD	1 to Many	user holds billboard licenses
USER	TAX	1 to Many	user is liable for land tax
USER	DISPUTE	1 to Many	user raises disputes
USER	AUDIT_LOG	1 to Many	user actions logged in audit
LAND	LAND_TRANSACTION	1 to Many	land has multiple transactions
LAND	LAND_HISTORY	1 to Many	land has multiple history events
LAND	LAND_SURVEY	1 to Many	land undergoes multiple surveys
LAND	INFRASTRUCTURE	1 to Many	land contains multiple assets
LAND	WORK_ORDER	1 to Many	land has multiple work orders
LAND	BILLBOARD	1 to Many	land hosts multiple billboards
LAND	TAX	1 to Many	land has yearly tax records
LAND	DISPUTE	1 to Many	land can be subject to disputes
LAND	DOCUMENT_REGISTRATION	1 to Many	land has multiple registered documents
INFRASTRUCTURE	MAINTENANCE	1 to Many	asset has multiple maintenance records

From entity	To entity	Cardinality	Relationship meaning
BILLBOARD	BILLBOARD_PAYMENT	1 to Many	billboard has multiple payment records

Section 3 — MySQL DDL commands

Copy and execute the following commands in sequence in your MySQL client. Tables must be created in the order shown to satisfy foreign key dependencies.

— Database creation

```
CREATE DATABASE IF NOT EXISTS land_management_db
    CHARACTER SET utf8mb4
    COLLATE utf8mb4_unicode_ci;

USE land_management_db;
```

— ADMIN table

```
CREATE TABLE admin (
    admin_id       INT AUTO_INCREMENT PRIMARY KEY,
    full_name      VARCHAR(100) NOT NULL,
    email          VARCHAR(150) NOT NULL UNIQUE,
    password_hash  VARCHAR(255) NOT NULL,
    role           ENUM('superadmin','officer','surveyor') NOT NULL,
    last_login     DATETIME DEFAULT NULL,
    is_active      TINYINT(1) DEFAULT 1,
    created_at     DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

— USER table

```
CREATE TABLE user (
    user_id        INT AUTO_INCREMENT PRIMARY KEY,
    full_name      VARCHAR(100) NOT NULL,
    nid_number     VARCHAR(30) NOT NULL UNIQUE,
    email          VARCHAR(150) DEFAULT NULL UNIQUE,
    phone          VARCHAR(20) NOT NULL,
    address        TEXT DEFAULT NULL,
    is_verified    TINYINT(1) DEFAULT 0,
    registered_at  DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

— LAND table

```
CREATE TABLE land (
    land_id                 INT AUTO_INCREMENT PRIMARY KEY,
    parcel_number           VARCHAR(50) NOT NULL UNIQUE,
    area_sqft               DECIMAL(12,2) NOT NULL,
    land_type               ENUM('residential','agricultural',
                                 'commercial','industrial','government') NOT NULL,
    mouza                   VARCHAR(100) NOT NULL,
    upazila                 VARCHAR(100) NOT NULL,
    district                VARCHAR(100) NOT NULL,
    current_status          ENUM('active','disputed','transferred','government')
```

```
                              DEFAULT 'active',
    current_estimated_value  DECIMAL(15,2) DEFAULT 0.00,
    registered_owner_id      INT NOT NULL,
    created_at               DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (registered_owner_id) REFERENCES user(user_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
);
```

— LAND_TRANSACTION table

```
CREATE TABLE land_transaction (
    transaction_id    INT AUTO_INCREMENT PRIMARY KEY,
    land_id           INT NOT NULL,
    from_user_id      INT NOT NULL,
    to_user_id        INT NOT NULL,
    transaction_type  ENUM('sale','inheritance','gift','court_order') NOT NULL,
    transaction_date  DATE NOT NULL,
    amount            DECIMAL(15,2) DEFAULT 0.00,
    deed_number       VARCHAR(100) NOT NULL UNIQUE,
    chain_valid       TINYINT(1) DEFAULT 1,
    approved_by       INT DEFAULT NULL,
    created_at        DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (land_id)      REFERENCES land(land_id) ON DELETE RESTRICT,
    FOREIGN KEY (from_user_id) REFERENCES user(user_id) ON DELETE RESTRICT,
    FOREIGN KEY (to_user_id)   REFERENCES user(user_id) ON DELETE RESTRICT,
    FOREIGN KEY (approved_by)  REFERENCES admin(admin_id) ON DELETE SET NULL
);
```

— LAND_HISTORY table

```
CREATE TABLE land_history (
    history_id         INT AUTO_INCREMENT PRIMARY KEY,
    land_id            INT NOT NULL,
    event_type         ENUM('survey','transaction','dispute_filed',
                           'status_change','billboard_placed','work_order') NOT
NULL,
    description        TEXT NOT NULL,
    event_date         DATE NOT NULL,
    recorded_by_admin  INT DEFAULT NULL,
    FOREIGN KEY (land_id)           REFERENCES land(land_id) ON DELETE RESTRICT,
    FOREIGN KEY (recorded_by_admin) REFERENCES admin(admin_id) ON DELETE SET NULL
);
```

— LAND_SURVEY table

```
CREATE TABLE land_survey (
    survey_id           INT AUTO_INCREMENT PRIMARY KEY,
    land_id             INT NOT NULL,
    survey_date         DATE NOT NULL,
    surveyor_admin_id   INT DEFAULT NULL,
    land_type_at_survey VARCHAR(50) NOT NULL,
    survey_method       ENUM('manual','digital','satellite') NOT NULL,
    notes               TEXT DEFAULT NULL,
```

```
    approved_by         INT DEFAULT NULL,
    FOREIGN KEY (land_id)           REFERENCES land(land_id) ON DELETE RESTRICT,
    FOREIGN KEY (surveyor_admin_id) REFERENCES admin(admin_id) ON DELETE SET NULL,
    FOREIGN KEY (approved_by)       REFERENCES admin(admin_id) ON DELETE SET NULL
);
```

— INFRASTRUCTURE table

```
CREATE TABLE infrastructure (
    asset_id          INT AUTO_INCREMENT PRIMARY KEY,
    land_id           INT NOT NULL,
    asset_name        VARCHAR(150) NOT NULL,
    asset_type        ENUM('building','road','drainage','bridge','utility') NOT
NULL,
    construction_date DATE DEFAULT NULL,
    condition_status  ENUM('excellent','good','fair','poor','critical') DEFAULT
'good',
    owner_entity      VARCHAR(100) DEFAULT NULL,
    FOREIGN KEY (land_id) REFERENCES land(land_id) ON DELETE RESTRICT
);
```

— MAINTENANCE table

```
CREATE TABLE maintenance (
    maintenance_id     INT AUTO_INCREMENT PRIMARY KEY,
    asset_id           INT NOT NULL,
    maintenance_date   DATE NOT NULL,
    maintenance_type   ENUM('routine','emergency','upgrade','demolition') NOT NULL,
    cost               DECIMAL(12,2) DEFAULT 0.00,
    assigned_admin_id  INT DEFAULT NULL,
    notes              TEXT DEFAULT NULL,
    next_scheduled_date DATE DEFAULT NULL,
    FOREIGN KEY (asset_id)          REFERENCES infrastructure(asset_id) ON DELETE
RESTRICT,
    FOREIGN KEY (assigned_admin_id) REFERENCES admin(admin_id) ON DELETE SET NULL
);
```

— WORK_ORDER table

```
CREATE TABLE work_order (
    work_order_id      INT AUTO_INCREMENT PRIMARY KEY,
    land_id            INT NOT NULL,
    work_type          ENUM('road_construction','survey','demolition',
                            'utility_install','inspection') NOT NULL,
    requested_by_user  INT DEFAULT NULL,
    assigned_admin     INT DEFAULT NULL,
    status             ENUM('pending','in_progress','completed','cancelled')
                       DEFAULT 'pending',
    start_date         DATE DEFAULT NULL,
    end_date           DATE DEFAULT NULL,
    notes              TEXT DEFAULT NULL,
    FOREIGN KEY (land_id)           REFERENCES land(land_id) ON DELETE RESTRICT,
    FOREIGN KEY (requested_by_user) REFERENCES user(user_id) ON DELETE SET NULL,
```

```
    FOREIGN KEY (assigned_admin)    REFERENCES admin(admin_id) ON DELETE SET NULL
);
```

— BILLBOARD table

```
CREATE TABLE billboard (
    billboard_id       INT AUTO_INCREMENT PRIMARY KEY,
    land_id            INT NOT NULL,
    owner_user_id      INT NOT NULL,
    license_number     VARCHAR(80) NOT NULL UNIQUE,
    size_sqft          DECIMAL(8,2) NOT NULL,
    installation_date  DATE NOT NULL,
    expiry_date        DATE NOT NULL,
    annual_fee         DECIMAL(10,2) NOT NULL,
    status             ENUM('active','expired','suspended','removed')
                       DEFAULT 'active',
    FOREIGN KEY (land_id)       REFERENCES land(land_id) ON DELETE RESTRICT,
    FOREIGN KEY (owner_user_id) REFERENCES user(user_id) ON DELETE RESTRICT
);
```

— BILLBOARD_PAYMENT table

```
CREATE TABLE billboard_payment (
    payment_id         INT AUTO_INCREMENT PRIMARY KEY,
    billboard_id       INT NOT NULL,
    payment_date       DATE NOT NULL,
    amount_paid        DECIMAL(10,2) NOT NULL,
    payment_method     ENUM('cash','bank_transfer','online') NOT NULL,
    status             ENUM('paid','pending','overdue') DEFAULT 'pending',
    received_by_admin  INT DEFAULT NULL,
    FOREIGN KEY (billboard_id)      REFERENCES billboard(billboard_id) ON DELETE
RESTRICT,
    FOREIGN KEY (received_by_admin) REFERENCES admin(admin_id) ON DELETE SET NULL
);
```

— TAX table

```
CREATE TABLE tax (
    tax_id          INT AUTO_INCREMENT PRIMARY KEY,
    land_id         INT NOT NULL,
    owner_user_id   INT NOT NULL,
    financial_year  VARCHAR(10) NOT NULL,
    assessed_value  DECIMAL(15,2) NOT NULL,
    tax_rate        DECIMAL(5,4) NOT NULL,
    tax_amount      DECIMAL(12,2) NOT NULL,
    paid_amount     DECIMAL(12,2) DEFAULT 0.00,
    due_date        DATE NOT NULL,
    status          ENUM('unpaid','partial','paid','overdue') DEFAULT 'unpaid',
    calculated_by   INT DEFAULT NULL,
    UNIQUE KEY uq_land_year (land_id, financial_year),
    FOREIGN KEY (land_id)       REFERENCES land(land_id) ON DELETE RESTRICT,
    FOREIGN KEY (owner_user_id) REFERENCES user(user_id) ON DELETE RESTRICT,
    FOREIGN KEY (calculated_by) REFERENCES admin(admin_id) ON DELETE SET NULL
```

```
);
```

— DISPUTE table

```
CREATE TABLE dispute (
    dispute_id        INT AUTO_INCREMENT PRIMARY KEY,
    land_id           INT NOT NULL,
    raised_by_user    INT NOT NULL,
    opponent_party    VARCHAR(150) DEFAULT NULL,
    dispute_type      ENUM('ownership','boundary','mutation',
                          'encroachment','tax') NOT NULL,
    status            ENUM('filed','under_review','resolved',
                          'escalated','dismissed') DEFAULT 'filed',
    escalation_level  TINYINT DEFAULT 1,
    filed_date        DATE NOT NULL,
    last_updated      DATETIME DEFAULT CURRENT_TIMESTAMP
                      ON UPDATE CURRENT_TIMESTAMP,
    resolution_date   DATE DEFAULT NULL,
    resolution_notes  TEXT DEFAULT NULL,
    handled_by_admin  INT DEFAULT NULL,
    FOREIGN KEY (land_id)          REFERENCES land(land_id) ON DELETE RESTRICT,
    FOREIGN KEY (raised_by_user)   REFERENCES user(user_id) ON DELETE RESTRICT,
    FOREIGN KEY (handled_by_admin) REFERENCES admin(admin_id) ON DELETE SET NULL
);
```

— DOCUMENT_REGISTRATION table

```
CREATE TABLE document_registration (
    document_id          INT AUTO_INCREMENT PRIMARY KEY,
    land_id              INT NOT NULL,
    document_type        ENUM('deed','survey_certificate','tax_clearance',
                             'court_order','mutation_order','other') NOT NULL,
    title                VARCHAR(200) NOT NULL,
    file_reference       VARCHAR(255) DEFAULT NULL,
    registration_date    DATE NOT NULL,
    expiry_date          DATE DEFAULT NULL,
    doc_status           ENUM('valid','expiring_soon','expired','revoked')
                         DEFAULT 'valid',
    verified             TINYINT(1) DEFAULT 0,
    registered_by_admin  INT DEFAULT NULL,
    FOREIGN KEY (land_id)             REFERENCES land(land_id) ON DELETE RESTRICT,
    FOREIGN KEY (registered_by_admin) REFERENCES admin(admin_id) ON DELETE SET NULL
);
```

— AUDIT_LOG table

```
CREATE TABLE audit_log (
    log_id        INT AUTO_INCREMENT PRIMARY KEY,
    table_name    VARCHAR(80) NOT NULL,
    operation     ENUM('INSERT','UPDATE','DELETE') NOT NULL,
    record_id     INT NOT NULL,
    admin_id      INT DEFAULT NULL,
    user_id       INT DEFAULT NULL,
```

```
    old_value     TEXT DEFAULT NULL,
    new_value     TEXT DEFAULT NULL,
    performed_at  DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (admin_id) REFERENCES admin(admin_id) ON DELETE SET NULL,
    FOREIGN KEY (user_id)  REFERENCES user(user_id) ON DELETE SET NULL
);
```

— Useful indexes

```
CREATE INDEX idx_land_district   ON land(district);
CREATE INDEX idx_land_status     ON land(current_status);
CREATE INDEX idx_land_owner      ON land(registered_owner_id);
CREATE INDEX idx_tx_date         ON land_transaction(transaction_date);
CREATE INDEX idx_tx_land         ON land_transaction(land_id);
CREATE INDEX idx_dispute_status  ON dispute(status, escalation_level);
CREATE INDEX idx_tax_year        ON tax(financial_year, status);
CREATE INDEX idx_doc_expiry      ON document_registration(expiry_date, doc_status);
CREATE INDEX idx_billboard_exp   ON billboard(expiry_date, status);
CREATE INDEX idx_audit_table     ON audit_log(table_name, performed_at);
```

— Trigger — auto-escalate disputes (30-day idle)

```
DELIMITER $$

CREATE EVENT evt_escalate_disputes
ON SCHEDULE EVERY 1 DAY
DO
BEGIN
    UPDATE dispute
    SET
        escalation_level = LEAST(escalation_level + 1, 3),
        status = 'escalated'
    WHERE
        status NOT IN ('resolved','dismissed')
        AND escalation_level < 3
        AND DATEDIFF(NOW(), last_updated) > 30;
END$$

DELIMITER ;
```

— Trigger — auto-update document status

```
DELIMITER $$

CREATE EVENT evt_update_doc_status
ON SCHEDULE EVERY 1 DAY
DO
BEGIN
    UPDATE document_registration
    SET doc_status = 'expired'
    WHERE expiry_date < CURDATE()
      AND doc_status NOT IN ('expired','revoked');
```

```
    UPDATE document_registration
    SET doc_status = 'expiring_soon'
    WHERE expiry_date BETWEEN CURDATE()
                          AND DATE_ADD(CURDATE(), INTERVAL 30 DAY)
      AND doc_status = 'valid';
END$$

DELIMITER ;
```

— Trigger — update land estimated value after transaction

```
DELIMITER $$

CREATE TRIGGER trg_update_land_value
AFTER INSERT ON land_transaction
FOR EACH ROW
BEGIN
    UPDATE land
    SET current_estimated_value = (
        SELECT AVG(amount)
        FROM (
            SELECT amount
            FROM land_transaction
            WHERE land_id = NEW.land_id
              AND amount > 0
            ORDER BY transaction_date DESC
            LIMIT 3
        ) AS recent
    )
    WHERE land_id = NEW.land_id;
END$$

DELIMITER ;
```

— View — overdue taxes

```
CREATE VIEW vw_overdue_taxes AS
SELECT
    t.tax_id,
    l.parcel_number,
    l.district,
    u.full_name  AS owner_name,
    u.phone,
    t.financial_year,
    t.tax_amount,
    t.paid_amount,
    (t.tax_amount - t.paid_amount) AS balance_due,
    t.due_date
FROM tax t
JOIN land l ON t.land_id = l.land_id
JOIN user u ON t.owner_user_id = u.user_id
WHERE t.status IN ('unpaid','overdue','partial')
  AND t.due_date < CURDATE()
```

```
ORDER BY balance_due DESC;
```

— View — active dispute summary

```
CREATE VIEW vw_active_disputes AS
SELECT
    d.dispute_id,
    l.parcel_number,
    u.full_name        AS raised_by,
    d.dispute_type,
    d.status,
    d.escalation_level,
    d.filed_date,
    DATEDIFF(NOW(), d.last_updated) AS days_since_update,
    a.full_name        AS assigned_officer
FROM dispute d
JOIN land l  ON d.land_id = l.land_id
JOIN user u  ON d.raised_by_user = u.user_id
LEFT JOIN admin a ON d.handled_by_admin = a.admin_id
WHERE d.status NOT IN ('resolved','dismissed')
ORDER BY d.escalation_level DESC, d.filed_date ASC;
```